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Effects of carbon-based fillers on thermal properties of fatty acids and their eutectics as phase change materials used for thermal energy storage: A Review

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ABSTRACT

Organic phase change materials (PCMs) are the most common heat storage components in latent heat based thermal energy storage (TES) systems. Among the different organic PCMs, fatty acids (FAs) have shown a real potential in important areas, such as, food and medical transportation, passive solar heating and cooling of buildings, etc., because of the suitable thermodynamic and kinetic properties. However, like all other organic PCMs, FAs also suffer from low thermal conductivity (TC) and sometimes seepage issues. Various conducting alien components (metal nanoparticles, metal oxides, etc.) and different carbon-based materials (expanded graphite, graphene, graphene oxide, carbon nanotubes, activated carbon, carbon nanofibers, and carbon black) have been evaluated extensively to enhance the TC. However, carbon-based materials were observed to be superior due to their excellent thermal conductivity, stability, lightweight, and lower cost. While analyzing the literature for this review article, it was found that the bulk of work focuses on FAs, and FAs based composite PCMs with carbon-based fillers and resulting PCMs demonstrated a good performance under the experimental conditions. No obvious discussion exists on the choice of carbon-based filler from a commercial point of view. So, this review attempts to critically evaluate the research work reported to date on the FAs and FA based composite PCMs, where carbon-based fillers have been used. This discussion is followed by comparative analysis to narrow the selection window for specific carbon filler in terms of the best performance.

1. Introduction

1.1. Phase Change Material (PCM) Based-Thermal Energy Storage (TES)

Increasing energy cost and the growing greenhouse gases in the atmosphere are pushing humanity towards suitable renewable energy sources. Higher utilization of renewable energy can positively affect our environment, improve energy efficiency, and at the same time, add economic value [1-3]. Most of the renewable energy sources are interment in nature. Therefore, a suitable storage is needed for its effective use. This applies to solar heat as well. It is a good source of renewable energy and is suitable for a variety of applications such as, space heating and cooling, hot water, even can be used to produce electricity, etc. However, due to its intermittent nature, a storage system is required. Among the different thermal energy storage (TES) technologies, phase change material (PCM) based latent heat storage system is the leader, due to its higher heat storage capacity, a wide range of temperature, cost-effectiveness, and eco-friendliness. A PCM supported TES can utilize and store heat energy simultaneously, and again, this stored energy can be used when the solar heat is unavailable (i.e., night time) [2,4]. The PCMs store and release the heat energy by changing its physical state at a particular temperature, which refers as latent heat [5,6]. The basic principle of this process is demonstrated in Fig. 1.

The PCM based TES systems have a wide range of application

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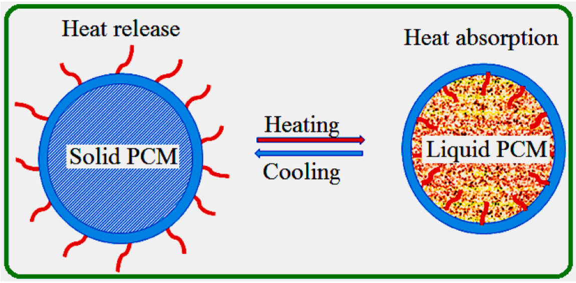

Fig. 1. Basic working principle of common solid-liquid PCMs.

possibilities, including heating and cooling of buildings, solar energy plants, solar drying devices, electricity generations, refrigerators, waste heat recovery, and residential hot water systems [7,8]. There is an increasing demand from various end-use practices, such as passive solar thermoregulation of building envelopes, food packs/transporters, textiles, electronic devices, etc. Inorganic PCMs have the largest market share at present. It is expected to grow primarily due to its slightly better thermal conductivity (TC) than organic PCMs and higher latent heat of fusion [1,9]. However, in recent years, organic PCMs are getting significant attention. Among the different applications, the construction of energy-efficient buildings is the largest market for PCMs, especially in Europe and North America, while it is growing in the Middle East due to the hot climate and increasing cost of space cooling [7,9]. Overall, PCMs have a substantial commercial utilization, and it is increasing as predicted by the leading market research organization [9].

The primary selection criterion of a PCM is to consider the appropriate phase transition temperature, as required for a specific application, along with its latent heat capacity. Other parameters, such as, chemical characteristics (non-flammability, no toxicity, long-term chemical stability, non-corrosive), physical properties (density, minimal vapor pressure, minimum volume transformation on phase change, congruent melting, TC), kinetic properties (substantial crystallization rate, lesser supercooling) and cost are also considered during the selection process [7,9]. All PCMs are broadly classified into three main classes, organic, inorganic, and eutectic (a combination of two or more natural, inorganic, and even both), as shown in Fig. 2 [10,11].

1.2. Fatty acids as PCMs

Among the different organic PCMs, fatty acids (FAs) have great potential in low-temperature TES systems [12,13,14]. Another advantage of FAs and their derivatives is that they can be acquired from both bio-based and synthetic sources. In recent years, extensive research has been carried out to isolate different FAs from feedstock and determine their suitability as bio-based organic PCMs. Most of these isolated organic compounds are primarily hydrogenated soy-wax or saturated fatty acid esters obtained from the hydrogenation of unsaturated FAs. Hydrogenation helps these organic compounds to stay stable throughout the phase change cycles without oxidation. The eutectic mixtures of FAs have also been developed by mixing different pure FAs to adjust the required melting temperature. The FAs have several favorable properties from the thermodynamics and kinetics angle, including reversible phase change behavior, a wide range of suitable melting/freezing temperature for different TES applications, good cycling, chemical, and thermal stability, non-toxicity, relatively low-cost, a small change in phase change vapor pressure, etc. [15-21]. As mentioned earlier, FAs suffer from leakage in different applications if not appropriately housed and have low thermal conductivity [22]. To resolve the seepage problem, typically, they are housed inside suitable porous support while different high conducting fillers have been employed to enhance their TC values. Among the many types of fillers, the carbon-based materials enable superior performance than the metal and metal oxides based fillers [10,23].

1.3. Carbon-based conductive fillers

Expanded graphite (ExG), carbon nanotubes (CNTs), graphene, graphene nanoplatelets (GNP), graphene oxide (GO), activated carbon (AC), carbon fiber (CF), and carbon black etc. as carbon-based materials have attracted great interest in many scientific fields due to their outstanding structural, thermal and electrical properties [24]. Among these materials, expand graphite (ExG) is one of the porous carbon materials with low density, high pore volume, and high TC. It can be prepared by heat treatment of acid treated-graphite at high temperature (~800 °C) [25]. Carbon nanotubes (CNTs) have a noticeable mono-dimensional cylindrical structure with high length-to-diameter ratios. The extraordinary TC value in the longitudinal direction can be as high as about 3000-3500 W/m K [26]. CNTs can be chemically functionalized by acid treatment to reduce long capped CNTs into

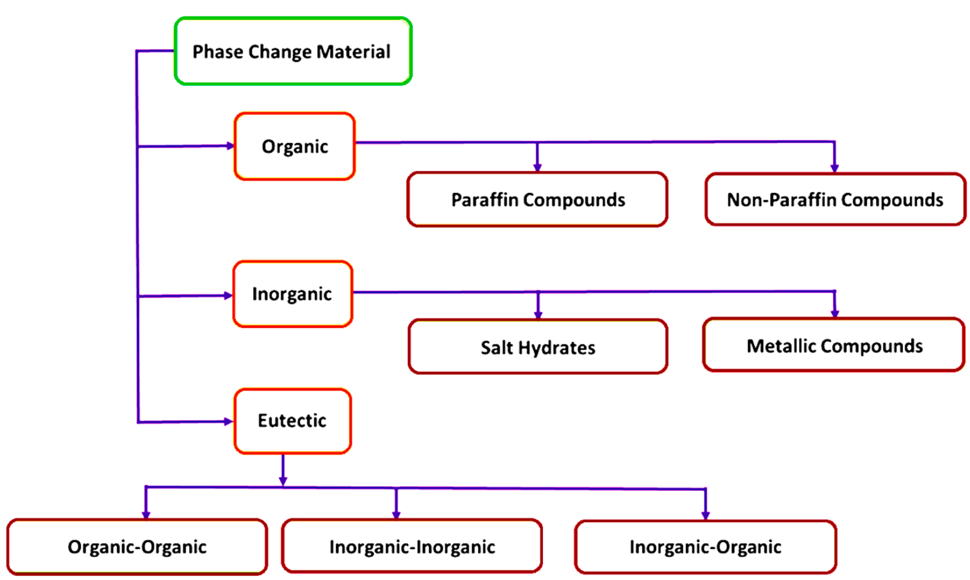

Fig. 2. Primary grouping of commonly used PCMs [10].

shorter ends-open pieces by adding oxygen functional groups to the surface of CNTs [27].

Graphene is a two-dimensional mono layered-carbon structure originated from the mechanically exfoliation of graphite and has an exceptionally high TC value of ~5000 W/m K [28]. A range of graphene nanosheets has been created by direct exfoliation of graphite [29] and also chemically or thermally reduced graphene sheets have been employed to enhance the TC of various organic materials. Moreover, the GO can be produced by oxidation of graphite [30] and its hydrophilic functional groups allow it to easily disperse in water, forming a colloidal solution. AC is a low-cost material and mostly produced from natural sources. So, it is a cost-effective alternative to CNTs, ExG, graphene, and GO and has been widely used in adsorption practices due to their pore structure and reasonable specific surface area [31]. CNFs are within the class of materials termed CNTs. They are at the nanometer scale and have notable thermal, electrical, electromagnetic shielding, and mechanical properties [32]. These properties make them popular additives for the production of composite materials [33]. Another carbon based filler material is the CB. It is commonly used as conductive filler because of its good inherent conductivity, relatively low cost and low density [34].

The carbon based-materials mentioned above have been investigated as attractive additives/fillers to enhance the thermal properties of several organic PCMs as well as in FAs and FA based PCMs because they have relatively low density, excellent thermal conductivity, mechanical durability, and photo absorbance ability. Additionally, these carbon materials are also environmentally friendly, non-toxic, economical, stable and chemically inert [35,36,37]. These materials have also preferred as effective support matrices in the fabrication of form-stable composite PCMs [38,39,40]; besides some of them have been used to increase thermal conductivity, photo-thermal conversion, and heat storage characteristics of phase change microcapsules slurries [41,42,43,44].

1.4. Aim of the present study

For different kinds of TES applications, stearic acid (SA), palmitic acid (PA), myristic acid (MA), lauric acid (LA), and capric acid (CA) as members of FAs are found to be the most studied PCMs. Like all other organic PCMs, these FAs have relatively low TC (in the range of 0.1-0.2 W/m.K), which hinders the heat flow and affects the heat storage/release time during operation. To overcome these issues, typically, conductive additives are used. Carbon-based materials received preference over metal additives due to their better dispersion in both solid and liquid state, low environmental impact, and sometimes provide shape-stability and lower cost. The effects of the major carbon-based materials, such as ExG, graphene, GO, CNTs, AC, CNF, and CB on TC and latent heat storage capacity (LHSC) and charging and discharging time during the operation of FAs and their eutectic PCMs were reviewed and comparatively discussed. Most of the researchers, who have worked mainly on FAs and FAs based composite PCMs with carbon-based fillers found their prepared PCMs demonstrating good performance under the experimental conditions. Therefore, it is difficult to find a conclusive opinion on a specific type of carbon base filler suitable for various applications, considering the dispersion issue, thermal properties, and even the cost and commercialization point of view. Keeping this fact in mind, we attempted herein to critically analyze the thermal properties of FAs, FA based eutectic PCMs and FAs included-CPCMs, where different carbon-based additives have been used, in terms of amount, incorporation techniques, and effects. Finally, an effort has been made to compare and highlight the better performing carbon fillers for FAs and their CPCMs that can be used frequently in the commercial-grade FA based PCMs.

2. Effect of carbon-based materials on TES properties and TC values of FAs included- PCMs

2.1. Expanded Graphite (ExG) incorporated FA-included PCMs

The ExG has been evaluated as a supporting matrix for form stabilize or shape stabilize composites with FAs, which prevented the seepage issue during melting [22,45]. Several groups have incorporated ExG in FAs, like MA [22], SA [46], LA [47], and PA [48], and obtained shape-stabilized composite PCMs (SSCPCMs) with improved thermal properties. A small amount of ExG significantly reduces the seepage issue in the melted state and restricts the comparative volume alteration of the PCMs depending on FA based eutectic throughout the phase change steps [49,50].

The CA, LA, and oleic acid (OA) have been used as organic PCMs due to their excellent properties like reversible phase change behavior, a wide range of suitable melting/freezing temperature for different TES applications, good cycling, chemical, and thermal stability, non-toxicity, relatively low-cost, and little change in phase change vapor pressure [15-21]. The surface morphologies of the ExG and the CPCMs prepared from CA-LA-OA and ExG were studied by scanning electron microscope (SEM), as shown in Fig. 3(a-h). It was found that the ExG exhibits a worm-like microstructure (Fig. 3a). Moreover, many pores with an interconnected network like structure at the micro-scale level have been observed (Fig. 3b), suggesting that the ExG offers a large surface area. It favors absorption properties. Conversely, due to the capillary forces, the influence of the volume alteration of CA-LA-OA throughout its solid-liquid transition process has been avoided. Fig. 3b shows the soft surface and layer like structure at the edge of the ExG. When CA-LA-OA is impregnated into the microspores of the raw ExG (Figs. 3c-f), the edge of each layer becomes thicker, and the surface roughness increases [22].

Additionally, the LHSC (109.18 J g^{-1}) and thermal conductivity (1.95 W m^{-1} K^{-1}) of SSCPCM3 was the best among all the SSCPCMs (Fig. 3g). The TC of SSPCMs declined with an increasing mass fraction of CA-LA-OA. However, the TC of SPCM7 (1.37 W m^{-1} K^{-1}) was 9.8 times higher than that of the pure ternary PCM mixture (0.14 W m^{-1} K^{-1}) (Fig. 3h).

A set of SA/benzamide(BA)/ExG composite PCM has been developed and studied [45]. The results obtained from the phase diagram and plot confirmed that the PCM included on composite has excellent thermal stability and consistency. The results suggested that the CPCM offers practical TES characteristics [45]. Binary eutectic mixture based LA-MA-ExG CPCMs showed better thermal stability than pure LA-MA eutectic mixtures [47]. A blend of CA-MA-PA and ExG CPCMs was prepared for their TES properties [47]. Fig. 4(a) demonstrates the DSC curves of ternary eutectic CA-MA-PA and ExG included CPCM. The DSC curves show that the melting temperature and the freezing temperature of the CA-MA-PA/ExG composite were relatively lower than that of the ternary eutectic CA-MA-PA.

Excellent TC of the ExG increases the heat transfer rate and helps to reduce the phase change temperature of the CPCMs. The melting and freezing latent heat values of the ternary eutectic CA-MA-PA/ExG were lower than the pure ternary eutectic mixture [47]. The change observed in the TC of the ternary mixture based CPCMs depending on the loading amount of ExG is presented in Fig. 4(b). It was determined that the inclusion of the ExG has markedly improved the TC of the ternary mixture. The TC of CA-MA-PA/ExG composites consisting of 5.0 wt% of ExG was 0.17 W $m^{-1}K^{-1}$, a 15% enhancement compared to a pure ternary eutectic mixture. Whereas, the 10.0 wt% ExG resulted in the TC of 0.18 W $m^{-1}K^{-1}$, a 20.8% enhancement of TC [47]. A CA-PA-SA/ExG was fabricated as CPCM (Fig. 5a,b), and the DSC analysis revealed that the melting temperature and LHSC of CA-PA-SA/ExG CPCM were 21.33 °C and 131.7 kJ kg^{-1}, respectively (Fig. 5c). The TC of all CPCMs round blocks was much higher than the pure ternary PCM and enhanced in a straight line depending on the change in density (Fig. 5d) [48].

Karaipekli et al. [49] improved SA's thermal properties by adding

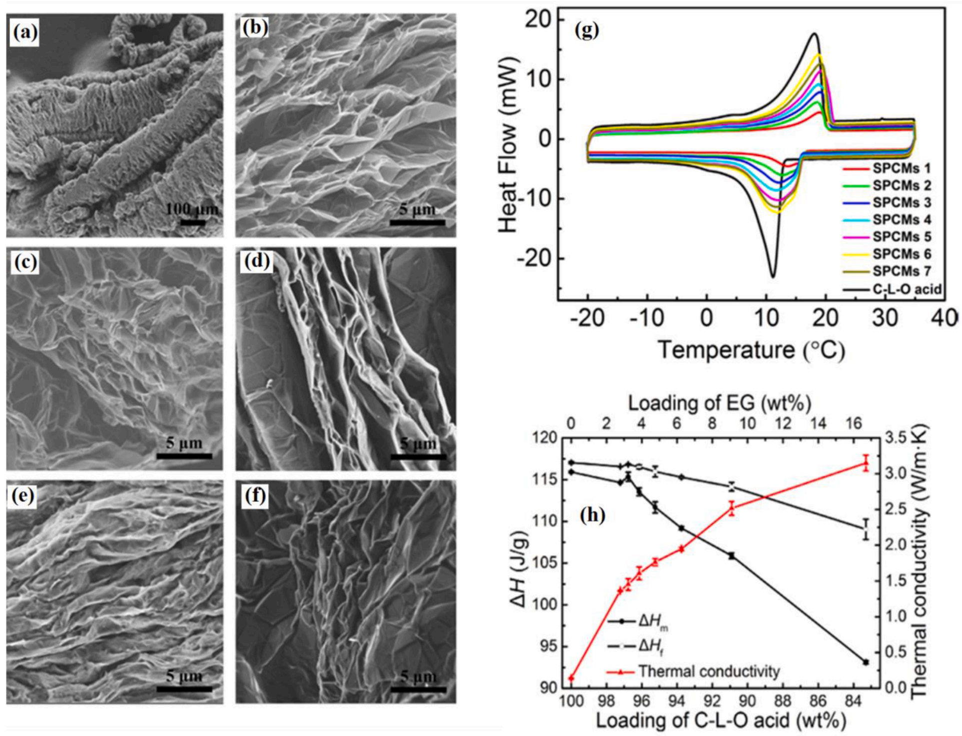

Fig. 3. SEM images of (a, b) raw ExG, (c) SSCPCM1, (d) SSCPCM3, (e) SSCPCPM5, and (f) SSCPCM7. (g) DSC results of the ternary mixture PCM and prepared CPCMs and (h) Effects of EG addition on LHSC and thermal conductivity of the CPCMs [22].

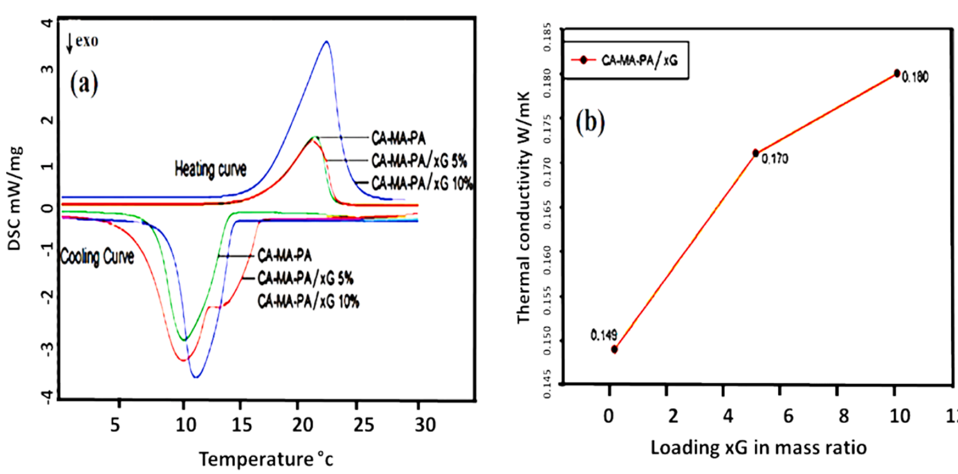

Fig. 4. (a) DSC curve of CA-MA-PA and the fabricated CPCM, (b) The change in the thermal conductivity of ternary mixture and CPCMs depending on the loading amount of ExG [47].

2-10 wt% of ExG. The morphological findings showed that the ExGs were homogeneously dispersed in the melted SA, resulting in exceptional physical and chemical properties of SA. Relative to pure SA, the TC of the CPCM with 10 wt% ExG was improved by 279.3%. The LHSC of pure SA and SA/ExG (90/10 wt%) were determined as 198.8 and 183.3 J g^{-1}, respectively. The LHSC of SA/ExG slightly reduced in comparison to that of pure SA. The study concluded that the fabricated CPCM had good potential for TES. Sari et al. [50] prepared the PA/ExG CPCM with good heat conduction property. When the mass fraction of ExG was 20 wt%, the CPCM showed no leakage during the heating period. Besides, the TC was 2.5 times higher than that of pure PA (0.17 W $m^{-1}K^{-1}$). Karaipekli and Sarı [51,52] separately produced two types of SSCPCMs, including CA-MA and LA, as PCMs and ExG as supporting material. The TC values of CA-MA/ ExG and LA/ ExG CPCMs were enhanced by ≈58% and 86% using 10 wt% ExG, respectively, while the thermal properties of the CPCMs were not considerably influenced. Consequently, both materials were proposed as promising CPCMs for different types of solar TES practices. Zhang et al. [53] fabricated LA-AM-PA/ExG CPCM (ternary PCM content: 94.7 wt%) with a phase change temperature of 31.41°C. The TC of the produced CPCM was much higher than that of the ternary PCM. Two different CPCMs were developed using two types of 5 wt% ExG (e.g., flake and amorphous) to the CA-MA-PA ternary mixture [54]. The LHSC of melting (143.7 J g^{-1}) and freezing (125.5 J g^{-1}) of CPCM, including ExG-F, changed and had a higher TC of 0.170 W m^{-1} K^{-1}, corresponding to an increase of about 114% compared to the pure ternary PCM. Accordingly, the CPCM with ExG-F had a better TES perspective, although ExG-A is comparatively cheaper than that of ExG-F. Additionally, other studies demonstrated

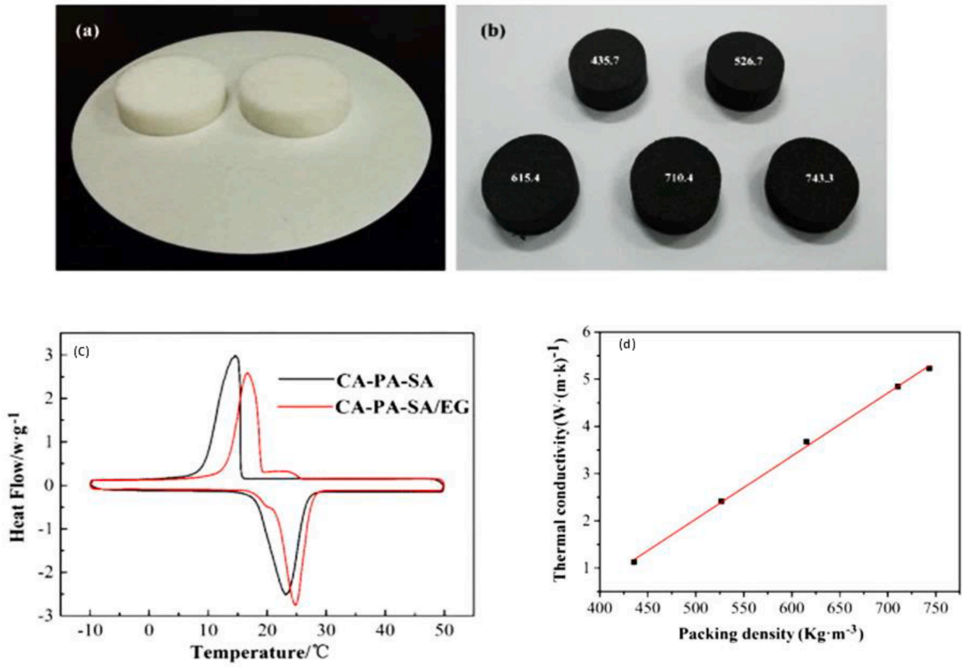

Fig. 5. (a) Pure CA-PA-SA ternary mixture, (b) CA-PA-SA/ExG composite, (c) DSC curves of the pure ternary mixture, and the prepared CPCM (d) Change in thermal conductivity of CPCM by increasing of packing density [48].

that the thermal properties of the FAs could be boosted by using ExG as an additive (Table 1).

2.2. Carbon nanotubes (CNTs) incorporated FA-included PCMs

The CNTs have been used extensively to enhance the thermal properties of different FAs, which include TC, heat storage/release time, and thermal stability [8,64]. In some cases, it also helped to control the seepage issue. The SA was mixed with -COOH functionalized CNTs in different weight ratios (1:1, 2:1, and 3:1) [65] and compared with pure CNTs based composite (Fig. 6a-d).

The composites with functionalized CNTs showed a sharp fall in the melting/freezing temperature (T_m/T_f) 30/22, 33/23, and 29/22.5 °C for the three composites compared to SA and composite with CNTs. The T_m/T_f reduction was attributed to the destructive interface imprisonment effect that arises due to the formation of the interface bonding between the -COOH groups of SA and a-CNTs, which were confirmed by FTIR spectral results. In the DSC curve of the composite-3 (Fig. 6 a-d), an additional peak appears at 42 °C, which is attributed to absorbed or free SA in the sample compared to the other three. Also, the produced CPCMs enhanced photo-thermal storage/release (Fig. 6e).

Several recent literature-based findings are discussed with critical

Table 1
Effects of ExG addition on the TES properties and TC values of some FAs and their CPCMs.

FAs/ExG or ExG/FA included-PCM	Latent heat of Melting ($J\ g^{-1}$)	Latent heat of Crystallization ($J\ g^{-1}$)	Observations	Ref.
CA-MA-PA/ExG	128.2	124.5	TC (3.67 $W\ m^{-1}\ K^{-1}$), thermal storage and release rates significantly increased as well as boosted thermal reliability	[47]
CA-PA-SA/ExG	131.7	127.2	TC was enhanced. Reduced the melting/cooling cycle time with excellent thermal cycling reliability even after 500 cycles	[48]
SA-Acetanitrile/ExG	220.0	176.1	Enhanced thermal conductivity, stability, LHSC (after 100 cycles only)	[55]
SA/Ultrathin graphite sheets	113.7	112.3	The best of the CPCM samples showed 10 times higher TC than pure SA. Good thermal stability below 180 °C and thermal reliability after 50 cycles	[56]
SA/Hollow porous Co_3O_4 -ExG	192.5	192.8	The best of the CPCMs samples showed nearly 7.67- times higher TC than that of pure SA in addition to good thermal stability and preventing the leakage issue of SA.	[57]
PA/polyvinyl butyral (PVB)/ExG	128.1	132.9	The TC of the best of CPCMs samples showed nearly 4.2 times higher than that of pure PA, besides higher latent heat storage capacity.	[58]
PA/PANI/ExG nanoplatelets	157.7	-	This work observed a 237% enhancement of the TC. However, as usual, the composites have lower heat storage capacity than the pure PA.	[59]
MA/graphite nanoplates composite	177.1	169.0	Improvement in TC of the CPCM as well as good repeated melting/freezing cycles.	[60]
CA-LA-OA/EG	114.6	116.6	TC of CPCM was 22.5 times higher, which caused better thermal stability compared to pure CA.	[22]
MA-PA-SA/EG	153.5	151.4	Proper phase change temperature, high latent heat, TC, and excellent thermal reliability after 1000 cycles.	[61]
LA-MA-SA/EG	137.1	131.3	Improved TC as well as good cycling TES reliability	[62]
CA/EG	132.6	134.2	Heat storage and release rates were boosted due to increased TC compared to pure FAs.	[63]
LA/EG	138.4	139.2	Results showed that the heat transfer rate of CPCMs was higher than that of pure FAs.	[63]
MA/EG	145.6	146.2	Heat charging and discharging rates of the CPCMs verified the improvement in TC. The heat transfer rate of CPCMs was higher than pure FAs.	[63]

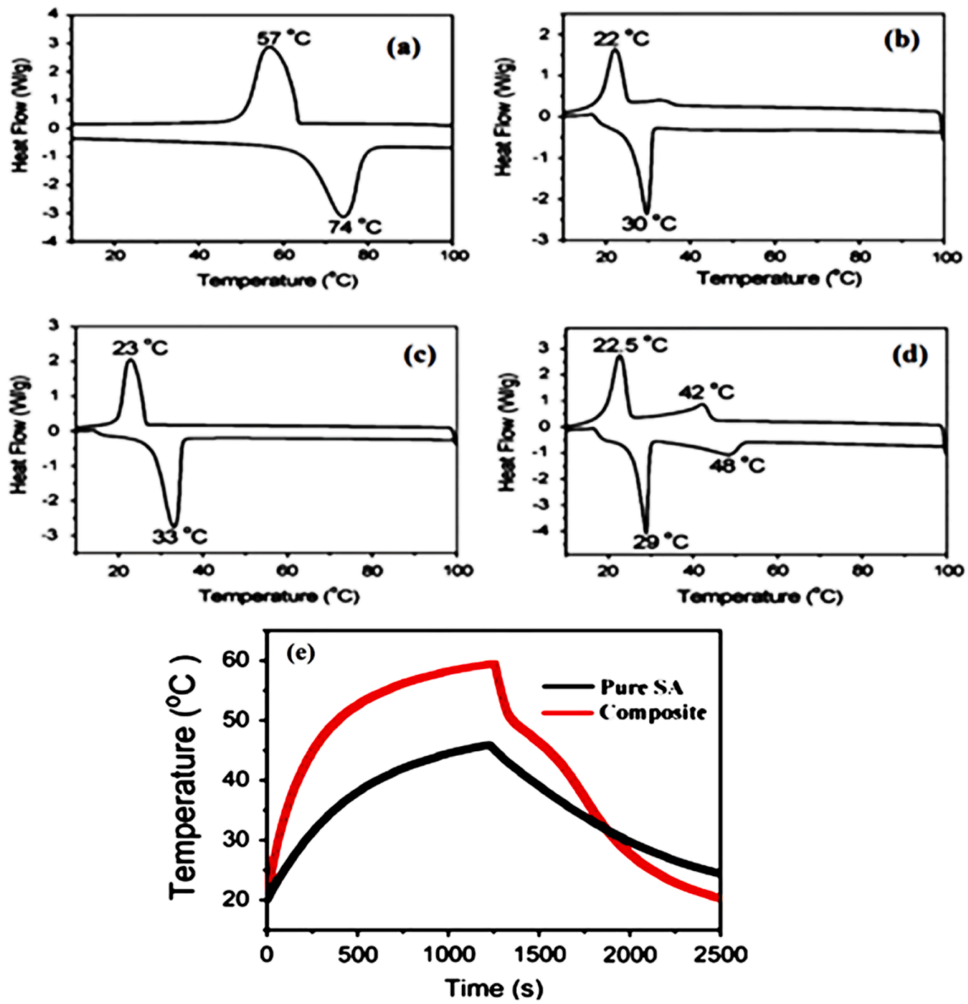

Fig. 6. DSC curves (melting-freezing) of (a) pure SA, (b) composite-1, (c) composite-2, and (d) composite-3, respectively; (e) photo-thermal storage/release performance curves of SA and composite-2 under simulated sunlight radiation [65].

analysis to highlight the impacts of CNT in FAs containing PCMs. Recently, Peng et al. [66] have prepared a new CPCM material using CA, fly ash (FA), and CNT. The CA was selected due to its suitable phase change temperature and good LHSC [66]. Initially, the mixing ratio of CA (20 wt%) was optimized with the help of the leakage test. Then a different mass fraction of CNTs was mixed into the CA/FA pre-composite using a hot plate at 60 °C to obtain SSCPCM. All components of the prepared SSCPCM are visible in the SEM image (Fig. 7a). Though there was visible agglomeration in the composites, CNTs were found to have distributed evenly. The heating/cooling time of CA/FA/CNT was shortened mainly due to the increased TC (Fig. 7 b,c).

Zhang et al. [67] investigated the effect of the CNTs on the thermal properties of the mixture of PA-SA. Different kinds of composites were prepared by changing the ratio of CNTs (5, 6, 7, and 8 wt%) in the mixture. The morphology of the obtained CPCM is depicted in the SEM image (Fig. 8a). The agglomeration of the CNTs and the FAs are visible in white color on the surface of the CNTs.

The results showed that the mixture's thermal conductivity were boosted by 20.2, 26.2, 26.2, and 29.7% for 5, 6, 7, and 8 wt% of CNTs, respectively. As shown in Fig. 8(b), the melting temperatures of the PA-SA/CNT composites are about 0.3-04 °C lower than that of the eutectic FA while the widths at half height of the composites are reduced in the range of 0.3-0.7 °C relative to that of the eutectic FA due to promoted heat transfer rate depending on the amount of added amount of the CNTs (in the range of 5-8 wt%). Conversely, as indicated in Fig. 8(c), compared to that of the eutectic PA-SA mixture, the freezing temperatures of the composites were increased by ~ 0.06-0.1 °C as the widths at half height of the DSC curves of the composites were diminished in the range of 0.3-0.7 °C in case of CNTs addition in mass fraction range of 5-8 wt%. Also, the FTIR spectral findings revealed that the silane chains were grafted onto the surface of O-CNTs.

Da et al. [68] used three different CNTs (pristine CNT, oxidized CNTs, and (γ-(2,3-epoxypropoxy) propyl trimethoxysilane grafted CNTs)) in 1:100 mass ratio of PA to prepare the CPCM, and studied their inherent properties. The SEM image of the PA/CNTs, PA/O-CNTs, and PA/G-CNTs composites are shown in Fig. 9 (left side: a-d), where an exact phase separation was observed in the CPCM sample with pristine CNTs and PA (Fig. 9b). Contrarily the other two samples with modified CNTs showed much better homogeneity and compatibility.

The PA/G-CNTs sample showed latent heat of 201.7 J g^{-1}, which was higher than the LHSC of pure PA (199.8 J g^{-1}). The high latent heat capacity could be a result of better adhesion between the G-CNTs and PA and contributed to the heat capacity of the γ-(2,3-epoxypropoxy) propyltrimethoxysilane [68]. The DSC curves of PA and CNTs, O-CNTs, and G-CNTs, are shown in Fig. 9 (right side: a,b,c). The PA/G-CNTs sample demonstrated enhanced stability compared to the other two, as no significant heat capacity was lost even after consecutive 100 cycles. On the other hand, the UV–Vis analysis results obtained for composites/ethanol solution demonstrated that a decrease in the surface energy and Van der Waals' force of CNTs, lead to weakening of agglomeration of CNTs. While the silane chains of G-CNTs reduces the surface energy and Van der Waals' force and therefore better dispersion were achieved in

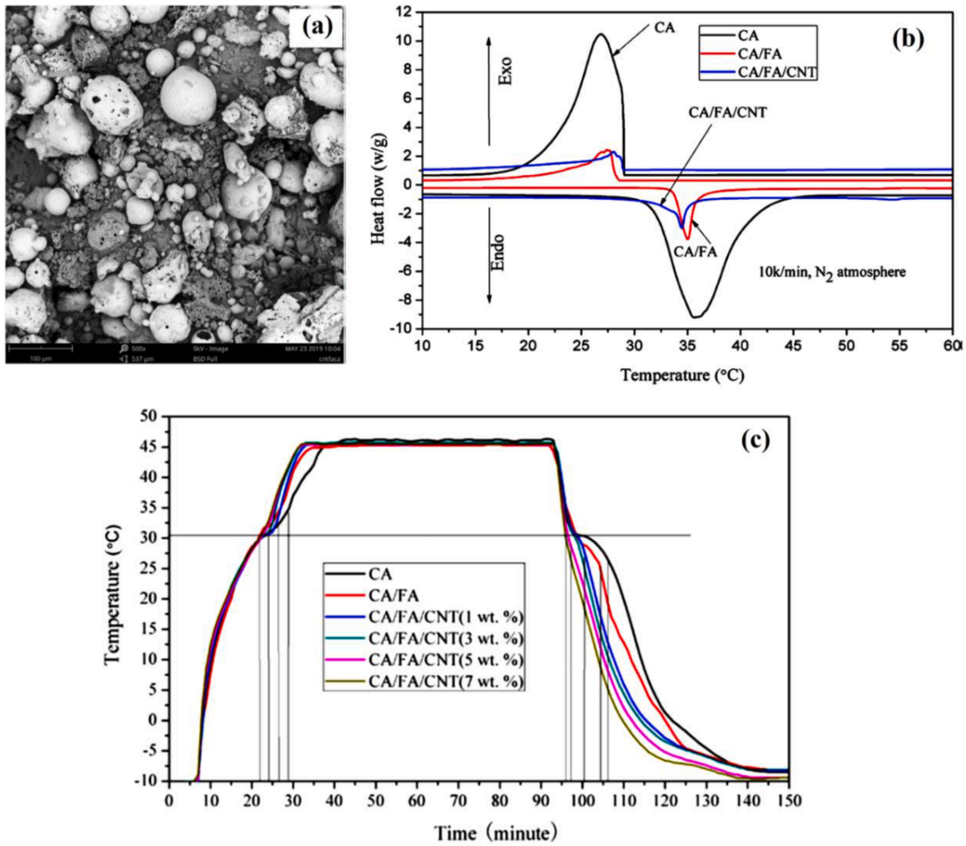

Fig. 7. (a) SEM images of CA/ FA/CNT, (b) DSC curves of CA, CA/FA, and CA/FA/CNT, (c) Cycling study curve of all the CPCM samples [66].

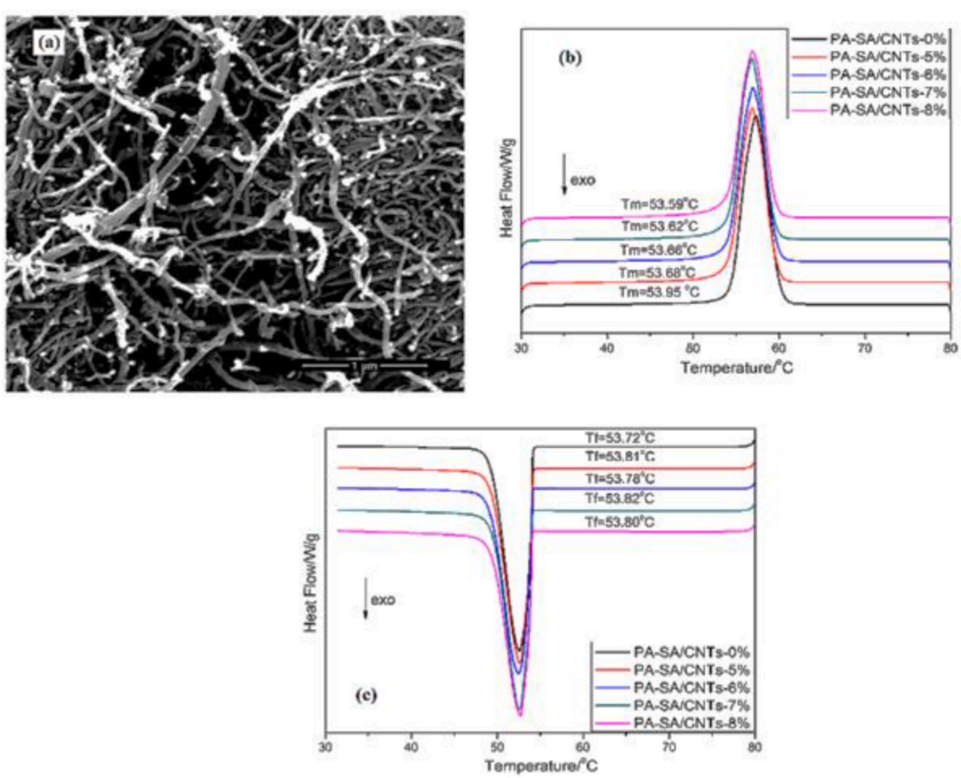

Fig. 8. (a) SEM image of CNTs mixed with the FA, DSC curves of (b) melting process, and (c) freezing process (PA-SA and CPCMs with various mass fractions of CNTs) [67].

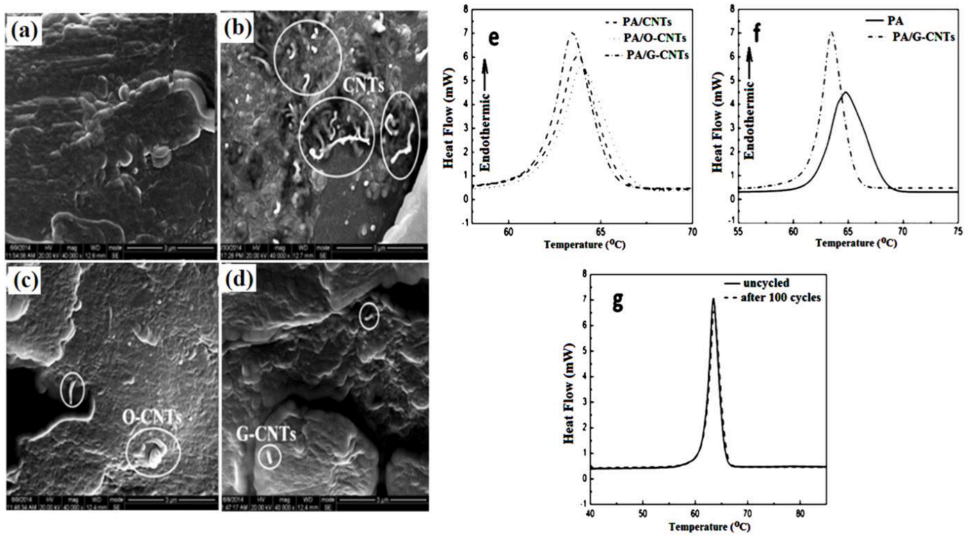

Fig. 9. At left side: SEM images of (a) PA; (b) PA/CNTs; (c) PA/O-CNTs; (d) PA/G-CNTs; and DSC curves of (e) CPCMs, (f) comparative DSC curve of pure PA and the champion CPCM sample and (g) DSC curve of the champion CPCM sample before and after 100 cycles [68].

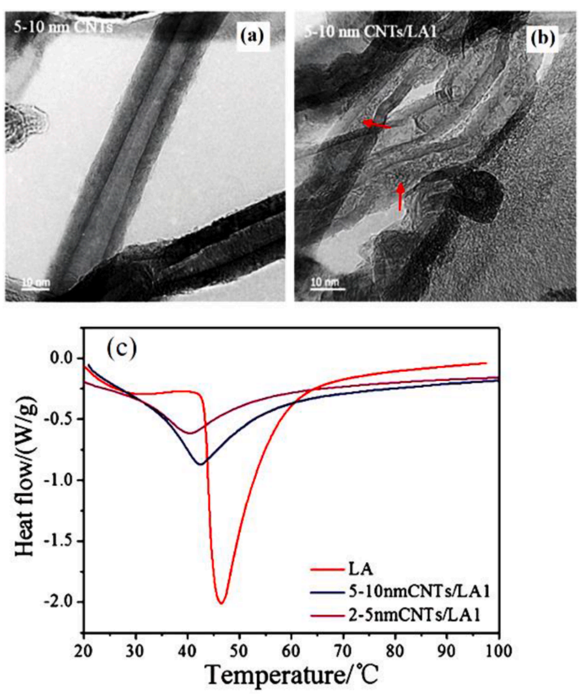

Fig. 10. TEM images of (a) CNTs and (b) CNTs/LA1 composites (the red arrow indicates the presence of the LA inside the CNTs), (c) DSC curves of pure LA, and CPCMs with 5–10 nm and 2-6 nm CNTs [69].

comparison to O-CNTs.

More recently, Feng et al. [69] filled CNTs (5 -10 nm and 2–5 nm diameters) with different LA weight ratios and compared their performances. The TEM image (Fig. 10a-b) confirmed LA's presence within the CNTs, and it was fully distributed within the channel.

Additionally, the melting temperature of the LA in the composite was 36.01 and 32.02 °C for the 5–10 nm CNTs/LA1 and 2-5 nm CNTs/LA1 respectively, which were lower than the melting temperature of pure LA (43.14 °C) (Fig. 10c). Again, at the same amount of LA loading, CPCM having CNTs with smaller diameter showed reduced melting temperature, and all the composites showed higher TC.

Meng et al. [70] prepared a eutectic mixture of LA, CA, and PA (3:5:2). The CPCM was prepared with the acid-treated CNTs (10 to 50 wt %) in acetone by ultrasonic vibration. The eutectic mixture had a melting temperature of 16.8 °C which decreased as CNTs loading increased. As observed in all the above-discussed literature, the melting enthalpy also reduced with the increasing amount of CNTs. However, the samples with 80 and 90 wt% of eutectic FA loading showed melting enthalpy of 101.6 and 122.3 J g^{-1}, respectively, which were closer to the melting enthalpy of the pure eutectic mixture (140.5 J g^{-1}). These two samples showed good thermal stability and enhanced TC. Wang et al. [71,72] composed PA/modified CNTs as CPCM *via* ultrasonic processing. The results showed that the TC of PA modified CNTs was increased by 51.6% as the LHSC of the resultant CPCM was decreased to 184.0 J g^{-1}. The same group [73] also investigated the effect of CNTs (with surface modified by oleylamine and octanol) on the phase change properties and TC of PA. Compared to the pure PA, the TC of modified-CNTs/PA CPCM was significantly improved. Simultaneously, the type of modifying agent grafted onto the surface of the CNTs chain substantially influenced the inter-molecular attraction within the composite PCMs. Accordingly, the phase change temperature of the CPCM was unchanged, and the latent heat decreased due to the hydrogen bonding between CNTs and –COOH group of PA.

Sarı et al. [74] developed silica fume (SF)/(CA-PA) as a novel CPCM with no leakage over its melting temperature. The hybrid structure of SF/CNTs maintained structural stability. It prevented the CA-PA leakage above its melting temperature due to the capillary and surface tension among the components (Fig. 11a-c).

The DSC curves indicated that the SSCPCMs with/without CNTs had the desired phase change temperatures (about 19–26 °C) and tolerable LHSC (about 46-49 J g^{-1}) for TES targets in energy effective-buildings (Fig. 11d-e). The TC of the prepared CPCM was increased by about 12.9, 32.3, and 56.1% by adding 1, 3, and 5 wt% of CNTs, respectively (Fig. 11f). Moreover, as shown in Fig. 11g, the CNT's addition significantly reduced the heating and cooling times of the SF/(CA-PA) because of enhanced TC. Additionally, the TES properties and TC enhancement ratio (%) of some pure FAs and FAs included-CPCMs doped with CNTs in different weight ratios (%) are presented in Table 2. As can be seen from the tabulated data, the LHSC of the FAs and their CPCMs are changed with the impregnation ratio of FA in the composite of and weight ratio of added CNTs. Higher impregnation of FA leads to higher LHSC for the FA comprised CPCMs. Moreover, increasing amount of CNTs slightly reduces the LHSC of FAs or FA comprised CPCMs.

On the other hand, as expected, the TC enhancement ratio of FA or its CPCM is boosted linearly depending on the increased weight ratio of CNTs. However, some of the studies show an adverse result for the TC

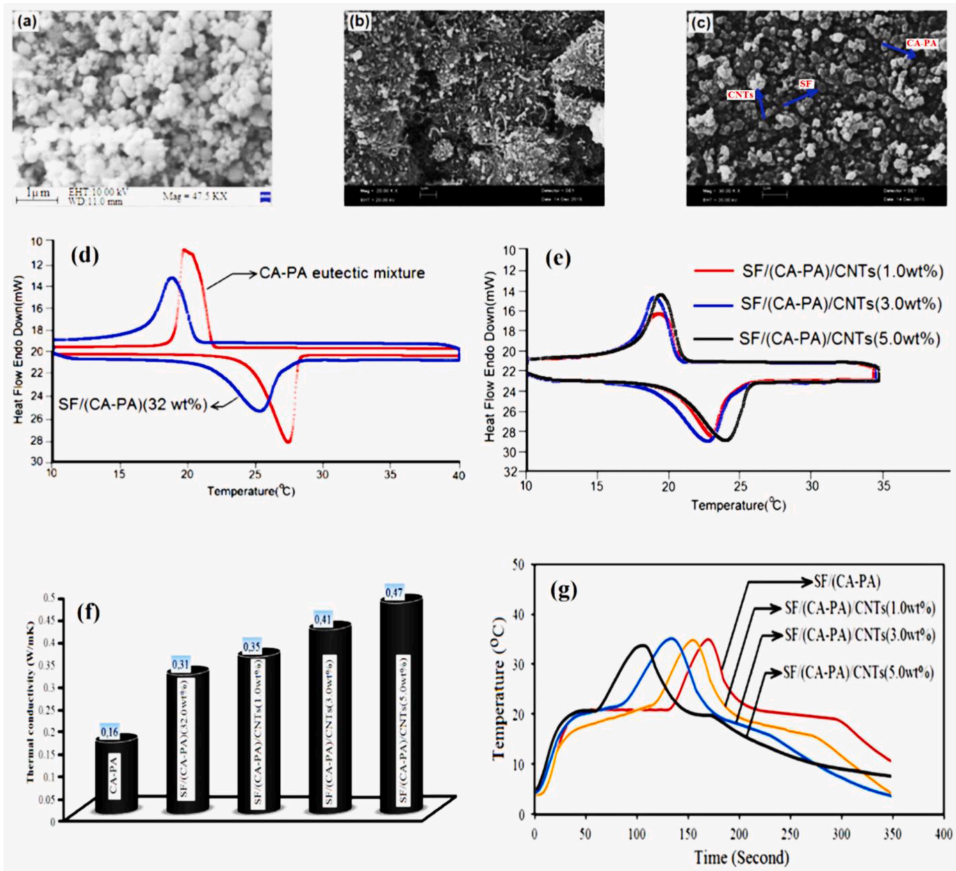

Fig. 11. SEM images of (a) SF, (b) CNTs, and (c) SF/(CA-PA)/CNTs; DSC curves of (d) pure CA-PA and SF/(CA-PA), (e) prepared SF/(CA-PA)/CNTs, (f) TC data of the prepared CPCMs with/without CNTs, and (g) heating and cooling curves of CPCMs with/without CNTs [74].

enhancement ratios of some FAs. For example, 1 wt% CNT addition to PA resulted in 46.0% enhancement in its while 20.2-29.7% enhancement was determined for TC value of PA-SA by addition of CNTs ratio of 5-8 wt%. It may be attributed to the purity/or chemical structures of used CNTs as well as the differences in TC measurement methods and conditions. It can also be understood from the tabulated results that the addition of CNTs to FAs entrapped in clay samples with very low thermal conductivity (about 0.04-0.08 W/mK) usually did not cause a high increase in the TC of FA incorporated-CPCMs as in pure FAs.

2.3. *Graphene nanoplatelets incorporated FA-included PCMs*

Excellent TC, ultra-lightweight, imperfection-free graphene sheets make it thermal conductive filler for FA included-CMs showed significantly enhanced TC and good LHSC. Harish et al. [79] prepared LA-included phase change nanocomposites using chemically functionalized graphene nanoplatelets. The introduction of only 1 vol% of graphene nanoplatelets improved the TC of LA by 230% without affecting the phase change enthalpy and melting temperature (Fig. 12a-b). The phase change enthalpy ($\triangle H_{fus}$) and the CPCM's melting temperature were found to be 181 ± 3 J g^{-1} and 43.8 ± 0.1 °C, respectively. The thermal interface conductance between graphene and PCM was ~100 MW m^2 K^{-1}, much higher than that of CNTs/PCM (Fig. 12b).

Mehrali et al. [80] developed PA/graphene nanoplatelets composite as leakage-free CPCM containing 91.94 wt% PA by the impregnation method. Graphene, with 300, 500, and 750 m^2 g^{-1} specific surface, were selected in this work. The LHSC of the CPCM, including graphene nanoplatelets with a surface area of 750 m^2 g^{-1}, was measured as 188.98 kJ kg^{-1}, which was close to that of pure PA (205.53 kJ kg^{-1}). The TC (2.11 W m^{-1} K^{-1}) of this CPCM was about 8 times to PA (0.29 W m^{-1} K^{-1}). As a result, this CPCM PA/GNPs with the shape-stabilized property was suggested for TES applications, such as power plants, solar collectors, and electronic devices [80]. The influence of adding

Table 2
The TES properties and TC enhancement ratio (%) of some pure FAs and their CPCMs doped with CNTs in different weight ratios.

FA/CNTs or FA included-CPCM/ CNTs	Melting temperature (°C)	Melting enthalpy of pure FA ($J\ g^{-1}$)	Melting enthalpy of FA included-CPCM/ CNTs ($J\ g^{-1}$)	Weight of CNTs (%)	TC enhancement (%)	Ref.
PA-SA/CNTs	53.6	177.7	163.0	5-8	20.2-29.7	[67]
CA-PA/SF/CNTs	25.4	151.5	39.12	5	56.1	[74]
PA/CNTs	60.1	208.0	184.0	1	46.0	[75]
SA/CNTs	68.1	210.9	176.2	10	119.2	[76]
LA/CNTs	39.7	198.2	16.81	10	185.5	[77]
CA/DT/CNTs	31.4	87.68	79.09	7	156	[78]

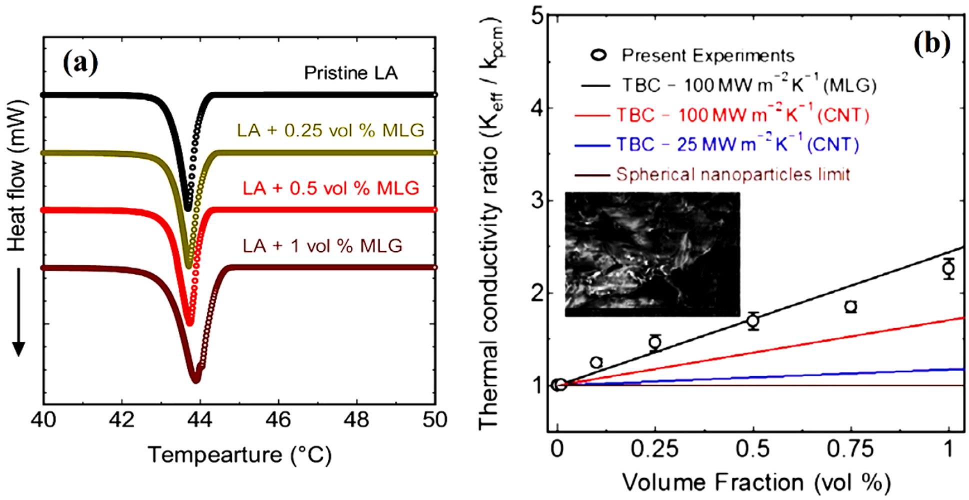

Fig. 12. (a) DSC melting curves of pure LA and the produced nano CPCMs, (b) The change in TC of LA depending on the volume fraction of added MLG [79].

nitrogen-doped graphene (NDG) at different fractions (1-5 wt%) on the thermal properties of PA was also studied [81]. The TEM images demonstrated (Fig. 13a-b) that NDG had a network structure, which allows PA adsorption. The capillary forces and surface tensions between the components resulted in excellent compatibility and enhancements in the TC. The DSC results show that the melting temperature increases with the increasing amount of NDG (Fig. 13c), but the LHSC value decreases.

The TC of PA was increased up to 6 times when 5 wt% of modified graphene was used (Fig. 13d); however, this insignificantly decreases the LHSC. Enhancement in the TC considerably reduced the heat storage/release time of the CPCMs (Fig. 13e). Both TC and the specific heat capacity of the CPCMs increased with increasing the amount of NDG (Fig. 13d, f). However, the sample with 97 wt% PA content and 3 wt% NDG showed shape stabilization, good LHSC of 199.48 kJ kg^{-1}, and thermal stability. This study concluded that this sample could be a potential material for the TES application satisfying latent heat capacity and leakage issue.

2.4. Graphene oxide (GO) incorporated FA-included PCMs

Akhiani et al. [82] used a one-step self-assembly strategy to create a CPCM using PA as well as oleylamine modified-reduced GO (Fig. 14a-b), eliminating the use of typical freeze-drying and impregnation stages. The hydrophobic van der Waals interaction between the oleylamine and PA aliphatic chains PA make easily adsorption in the modified-GO network. Also, the modified-GO network facilitated the crystallization of PA with an improved heat transfer and an outstanding shape-stability. Due to the use of the minimum amount of GO, the LHSC of PA was not affected, and the CPCMs exhibited slightly higher LHSC even after long thermal cycling (202 J g^{-1}) (Fig. 14c) [82]. The TC was increased exponentially with the increasing amount of OA-rGO, as shown in Fig. 14d. Furthermore, the CPCM's cooling rate containing 2 wt% GO was higher than the pure PA because of the improved TC of the CPCMs (Fig. 14e). According to the results, the CPCM containing 2 wt% GO released good light-to-heat transduction performance compare to the pristine PA.

In a similar approach, Li et al. [83] produced SSCPCM of SA/GO by housing SA in the multilayer GO's interlayer spaces. The thermal properties of SA/GO composite are found to be relative to the mass ratio of SA:GO. The FTIR spectroscopy analysis results confirmed the presence of some interface interactions between the -COOH group of SA and the -OH groups in GO. The developed SA/GO composite displayed significantly reduced melting (T_m) and freezing point (T_f) compared to pristine SA duo to interface interactions confirmed by FTIR analysis. The LHSC was determined as 55.7 J g^{-1}, which corresponds to the TES capability

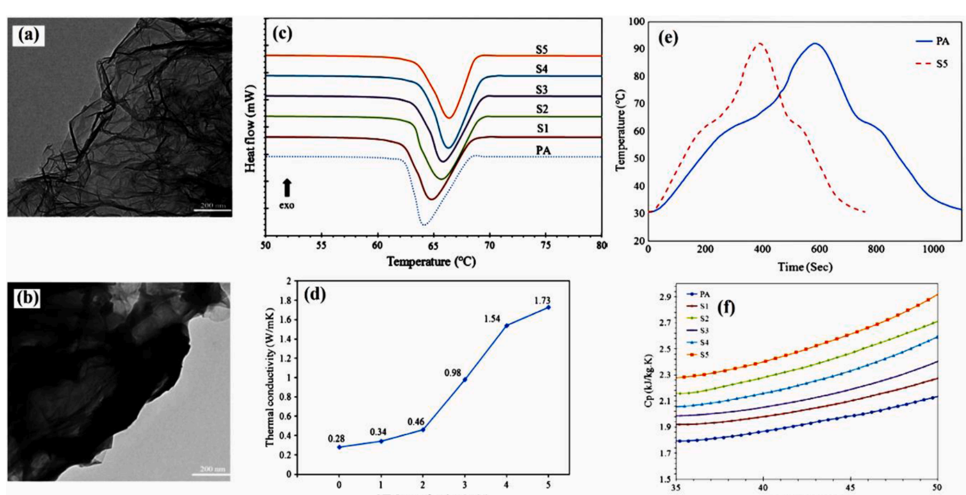

Fig. 13. (a-b) TEM images of NDG and PA/NDG; (c) DSC curves of the fabricated CPCMs, (d) Thermal conductivity of the CPCMs with the amount of added NDG, (e) Total heat charging and discharging times of the CPCMs, and (f) Specific heat capacity of CPCM with a change in temperature [81].

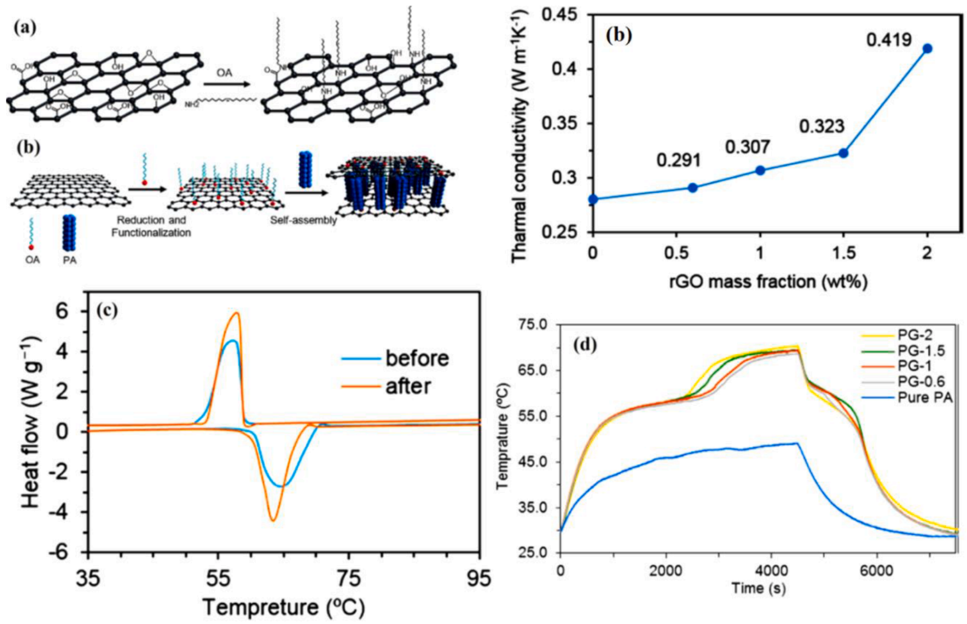

Fig. 14. (a) GO functionalization process, (b) The change in thermal conductivity of PA with the increase of a mass fraction of GO, (c) DSC curves of the CPCM containing 2 wt% GO after and before thermal cycling, (d) heat storage/release curve of pure PA and the CPCMs [82].

rate of 82.4%. It could be due to the lower amount of SA loading within the GO. In the sample-3, the loaded amount of SA was nearly 70%, which eventually showed better thermal storage capacity than the other two samples and almost close to that of pure SA. For all the studied composite samples, the graphene shell provided improved shape stability to the composite throughout the phase change process and enhancement of the TC [83]. The PA/GO composite was prepared *via* vacuum impregnation [84]. The TC (1.02 W m^{-1} K^{-1}) of CPCMs was enhanced noticeably by the addition of GO. The phase change temperatures and LHSC of the developed CPCMs were determined as 60.45 °C and 101.23 kJ kg^{-1}, respectively [84].

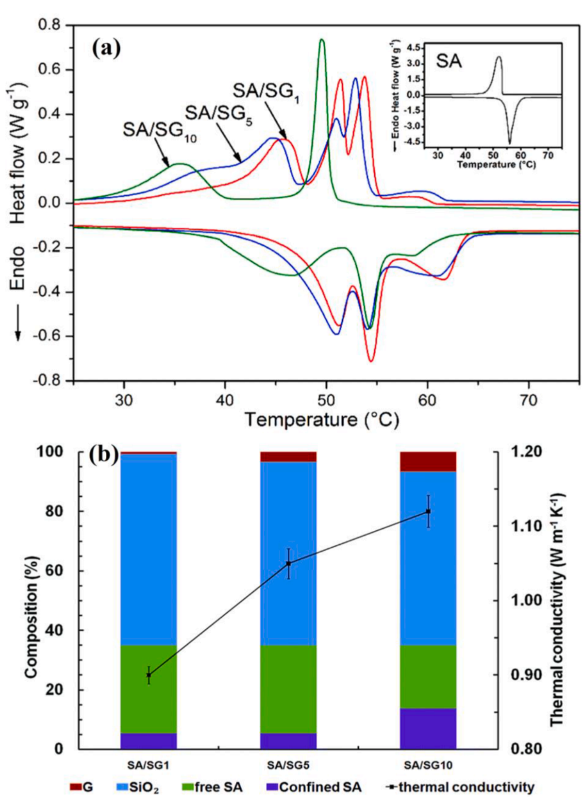

Fig. 15. (a) DSC curves of the prepared graphene-decorated silica (SG)/SA CPCMs, (b) The change in TC depending on the component composition of SG/SA CPCMs [86].

Song et al. [85] prepared GO/carbon nano-felts (RGO/CNFs) using the carbonization of GO loaded amidoxime surface-functionalized polyacrylonitrile (ASFPAN) nano-felts at different temperatures, and only 2 wt% of GO loaded on the ASFPAN was achieved. An LA-MA-SA ternary eutectic mixture was mixed with the RGO/CNFs-1, RGO/CNFs-2, and RGO/CNFs-3, respectively, to make SSPCM-1, SSPCM-2, and SSPCM-3 samples. The composite PCM, FSPCM-3, was prepared with RGO/CNFs-3 (carbonized at 950 °C) and the eutectic mixture showed a better loading of the eutectic mixture (73%) and a higher TC of 1.88 W m^{-1} K^{-1}. The TC was nearly 583% higher than the pure eutectic mixture [85].

Li et al. [86] prepared graphene-decorated silica (SG)/SA CPCMs with good thermal stability and higher TC. With 34.3% of SA, the best CPCM sample showed an effective LHSC capacity per unit mass (149.6 J g^{-1}). Higher graphene-decorated silica loading reduced the melting and freezing temperature (Fig. 15a).

As observed in all previous reports, addition of filler reduces the LHSC due to the reduced amount of SA loading in the CPCMs. Fig. 15b showed that the SA/SG_5 sample had a very similar composition than the SA/SG_1 but higher TC.

2.5. Activated carbon (AC) incorporated FA-included PCMs

The ACs have also been used to adjust the properties of the FAs included-PCMs. Wan et al. [87] prepared carbon as a bio-char from a pinecone and used as a supporting material to develop PA/bio-char CPCM. Homogeneous mixing became visible in the SEM images of the composite PCM samples (Fig. 16a-d). The champion sample (PA/PB-4) showed the highest melting and freezing enthalpy of 84.74 and 83.81 kJ kg^{-1}, respectively. However, these values were much lower than the melting and freezing enthalpy of pure PA, determined as 219.63 and 213.91 kJ kg^{-1}, respectively. In addition, it was concluded that the CPCM has a good thermal cycling stability (Fig. 16e) and exhibits an enhanced TC of 0.3926 W $m^{-1}K^{-1}$, surprisingly, which was even higher

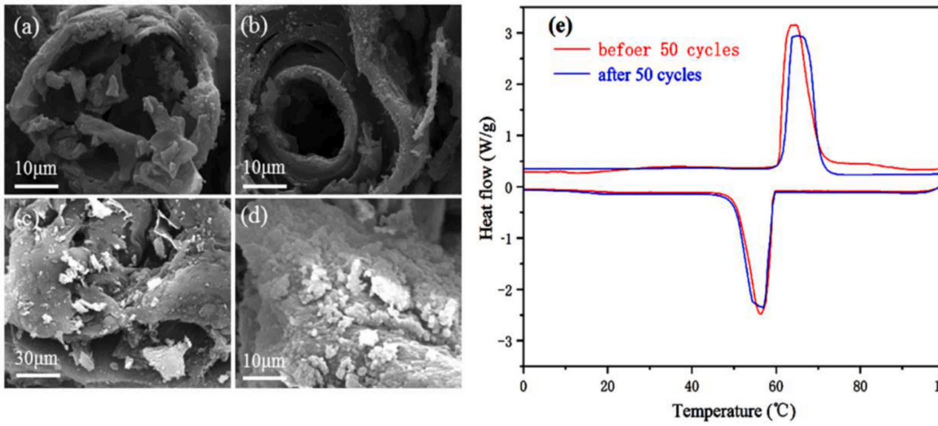

Fig. 16. SEM image of (a, b) bio-char, (c, d) developed CPCM, and (e) DSC curve of the champion (PA/PB-4) sample before and after 50 cycles [87].

than that of the bio-char. The bio-char and high surface area and porosity prevented the leakage of the PA in the champion sample (PA/PB-4).

A ternary eutectic mixture was prepared using CA, PA, and SA, which showed melting and freezing temperatures of 19.93 and 16.84 °C, respectively [88]. In this work, activated carbon was used to improve the thermal properties of the eutectic PCM mixture. The authors found the LHSC value of 25.80, 34.62, 46.54, 67.15 J g^{-1} for the CPCM sample PCM1 (50:50), PCM2 (55:45), PCM3 (60:40) and PCM4 (65:35), respectively, much lower than the pure eutectic sample (129.4 J g^{-1}). The addition of activated carbon led to a decrease in the melting temperature and an increase in the freezing temperature of the CA-PA-SA/AC. However, the thermal storage-release time was faster than the pure eutectic PCM [88]. Similarly, Chen et al. [89] prepared LA/AC CPCMs at different mass ratios of 1:5, 1:4, 1:3, and 1:2 and shape stabilization observed at 33.3 wt% LA loading. The TC of the LA/AC (1:2) was increased by 46.7% with the addition of just 1 wt% of AC. The DSC curves demonstrated that the melting temperature and LHSC of the CPCMs were boosted by increasing the amount of LA. The CPCM sample with 33.3 wt% LA had a melting temperature and LHSC of 44.07 °C and 65.14 J g^{-1}, respectively. Hussain et al. [90] enhanced the TC of liquid CA-OA eutectic PCM by 12-55% with the addition of AC in the range of 0.02-0.1 wt%, while its degree of sub-cooling was reduced to 30 - 63% relative to the eutectic PCM. An addition of 0.1% AC eliminated the sub-cooling effect. Moreover, the reduction in the LHSC of the eutectic PCM was caused by physicochemical changes.

2.6. Carbon nanofiber (CNF) incorporated FA-included PCMs

Karaipekli et al. [49] improved the thermal properties of melted SA by using CNF additive in mass fractions of 2, 4, 7, and 10%. The TC of the CPCM was increased linearly with the increasing amount of CNF, and the champion sample with 10 wt% of the CNF showed 217.2% enhancement of TC value. However, the LHSC of SA/CNF (90/10 wt%) (184.6 J g^{-1}) was found to be 7% lower than the LHSC value of the pure PA (198.8 J g^{-1}). Cui et al. [91] submerged the eutectic mixtures CA-LA, CA-PA, and CA-SA to polyacrylonitrile (PAN) and carbon nanofiber mats. The eutectic-PAN nanofiber mats exhibited better impregnation capability (more than 90%), whereas CNF mats had 81.3% absorption. However, CNF mats based CPCMs displayed 49.5% shorter melting/-freezing times than the PAN-based composites since the CNF mats had relatively higher TC value compared to the PAN nanofibers.

2.7. Carbon black (CB) incorporated FA-included PCMs

The CB is a low-cost conductive filler compared with CNTs and graphene and has extensively used in the production of polymer composites [92] because of its good characteristics such as large specific surface area and nanoscale-particle size. Also, the high TC makes CB nanoparticles promising additive agents for the enhancement of thermal properties of an organic PCM [93]. Moreover, the CB was used as a carrier matrix for shape-stabilization as well as a filler to increase thermal stability and TC value of CA-SA eutectic mixture. The TC enhancement ratio of the studied FA eutectic mixture was determined as

Table 3
Comparison of some carbon-based additives on latent heat storage capacity (LHSC) of some FAs.

FA	LHSC of melting/solidifying (J g^{-1})			Ref.
	ExG[1]	CNTs[2]	Graphene[3]	
CA	LHSC of pure CA:172.42/174.11	LHSC of pure CA:163.37	LHSC of CA:187.7/190.1	[1] [63]
	LHSC of sample: 132.64/134.21	LHSC of sample: 79.09/79.11	LHSC of sample: 106.2/ 108.6	[2] [78]
	Sample: LA/ExG (20 wt%)	Sample: 54% CA/46% DT/7% CNTs	Sample: CA (60 wt%) CA/GO-DI-2	[3] [97]
LA	LHSC of LA:178.14/181.63	LHSC of LA: 175.0	LHSC of LA:180/n.a	[1] [62]
	LHSC of sample: 138.43/139.16	LHSC of sample: 54.6/n.a	LHSC of sample: 181/n.a	[2] [69]
	Sample: LA/ExG (20 wt%)	Sample: CNTs/54.8 % LA	Sample: LA/1.0 v/v% MLG	[3] [83]
				[2,3] [96]
MA	LHSC of MA:194.90195.15	LHSC of MA:191.90/195.15	LHSC of MA:194.90/195.15	[1,2,3] [97]
	LHSC of sample: MA:191.27/191.15	LHSC of sample:191.99/191.19	LHSC of sample:189.18/190.88	
	Sample: LA/nanoExG (1 wt%)	Sample: LA/MWCNTs (1 wt%)	Sample: LA/GNPs (1 wt%)	
PA	LHSC of PA:194.45/192.59	LHSC of PA:199.8	LHSC of PA:205.53/ 209.42	[1] [50]
	LHSC of sample: 148.36/149.66	LHSC of sample: 187.1/n.a	LHSC of sample: 188.98/ 191.23	[2] [98]
	Sample: PA/ExG (20 wt%)	Sample: PA/CNTs (1.0 wt%)	Sample: including 5 wt% of GNPs	[3] [84]
SA	LHSC of SA: 219.39/ 216.65	LHSC of SA: 219.39/ 216.65	LHSC of SA: 177/173.8	[1] [93]
	LHSC of sample: 191.93/ 194.76	LHSC of sample: 178.49/175.54	LHSC of sample: 51.32/ 49.18	[2] [96]
	Sample: SA/CNTs (9 wt%)	Sample: SA/CNTs (9 wt%)	Sample: SA(34.3 wt%)/SG5)	[3] [99]

Table 4
Comparison of the TC enhancement ratio (%) for some. FAs including different kind of carbon-based materials.

FAs	Enhancement (%) in TC			References
	ExG[1]	CNT[2]	Graphene[3]	
CA	-	31.3	292.6	[1] [63] [2] [78] [3] [87]
MA	15.2	33.7	101.5	[1,2,3] [73]
LA	-	72.0	230.0	[1] [63] [2] [69] [3] [73]
PA	73.0	30.4	627.0	[1] [50] [2] [88] [3] [84]
SA	515.9	42.0	303.0	[1] [96] [2] [96] [3] [99]

about 31% in the case of 21.60 wt% CB matrix usage [94]. In another study, Mishra et al [95] prepared CB loaded LA-based SSCPCM. The findings showed that the TC was enhanced by ~195% with the addition of 3.5 wt% of CB nano particles. The improvement in TC was associated with interconnected percolation networks to be created throughout the solidification of LA as well as superior volume filling capability and compressibility of CB nano particles.

2.8. Comparison of the effect of some carbon-based additives on LHSC and TC of some FAs

Tables 3 and 4 present a comparison to conclude the impact of the different carbon-based additives on LHSC and TC enhancement (%) of the FA-included CPCMs and predict the better additive from the literature analysis.

As observed in Table 3, the LHSC of the carbon-based material-included FAs are varied depending on the different parameters such as amount and phase change enthalpy of FAs, type and amount of additives, experimental measurement conditions, etc. By considering all these factors, it can be concluded that the graphene could be a better choice because of its layered structure. The layered structure of graphene can confine more PCM and can even prevent the leakage [90,91]. On the other hand, by analyzing the tabulated results given in Table 4, it can be conluded that ExG, CNT, and graphene are able to imrpove the TC of SA, LA, and PA. However, graphene is found to be more effective to to enhance the TC of the examined FAs.

3. Conclusion and Future prospects

The latent heat storage technologies using FAs or their CPCMs have resulted in different types of TES applications in recent years. However, relatively low TC (0.1-0.2 W/m.K) leads to a temperature reduction and thus their melting/solidification rates cannot reach their desired intensity. With this regard, incorporating carbon-based conductive fillers effectively improve the TC of the FAs or their CPCMs and enhances the heat transfer rate in the overall TES system. Carbon-based materials, such as ExG, Graphene, GO, CNT, CNF, AC and CB have been found to be effective to boost the TC of FAs, and their composites, which have yielded promising results. The detailed literature investigation showed that,

(i) The enhancement ratio in TC of their CPCMs is strongly dependent on the type, chemical structure (modified or unmodified), density, aspect ratio, surface area, doping amount, and TC value of the carbon based-materials.
(ii) The LHSC of FAs or their CPCMs is somewhat reduced with the weight amount of carbon based-additive.
(iii) The addition of CNTs to FAs entrapped in clay samples with very low thermal conductivity did not cause a significant enhancement of the TC value of the FA incorporated-CPCMs as in the case of pure FAs.
(iv) Doping over a certain amount with carbon-based material can cause agglomeration problems in pure FA or its CPCM. However, this problem can be circumvented through modified carbon-based material largely due to their ability to yield a homogeneous dispersion state.
(v) The enhanced TC results in significantly reduction in melting and freezing time of FAs and their CPCMs as well as a remarkable improvement in heat transfer and thermal performance of TES system.
(vi) Doping FAs or FA incorporated CPCMs with small amounts of carbon based-material considerably help to overcome the seepage issue in the melted state during the production of the composite PCMs.

The present review demonstrates that carbon structures are always a better choice than those of metal or metal oxide nanoparticle based additives due to their lower cost, better dispersion in the solid and liquid state, low environmental impact, and sometimes provide shape-stability. After comparison, it was found out that graphene fillers showed better dispersion, better retention of the heat storage enthalpy, and better TC than other carbon-based materials (ExG, CNT, etc.). In some cases, the graphene incorporated CPCMs showed similar or even slightly higher latent heat of enthalpy than the pure FA. Again, CNT and GO have a higher price tag than ExG; considering this point, ExG can be another suitable choice for cost-effective applications and where a higher amount of filler is required. All these fillers have an issue with the dispersion within the FAs, but pre-treatment (functionalization) could enable better dispersion. Overall, all carbon-based fillers increase TC significantly. With other important factors, for example thermal energy storage enthalpy, dispersion, dispersion stability and cost, it can be concluded that a lot of work is still required, and application-specific selection of fillers will be a systematic and balanced approach.

Declaration of Competing Interest

None.

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